13.1 EMF INDUCED IN A COIL ROTATING IN A MAGNETIC FIELD

Consider a rectangular coil of N turns and of length a and width b rotating with uniform angular velocity ω about its axis in a uniform magnetic field **B** (Fig. 13.1).

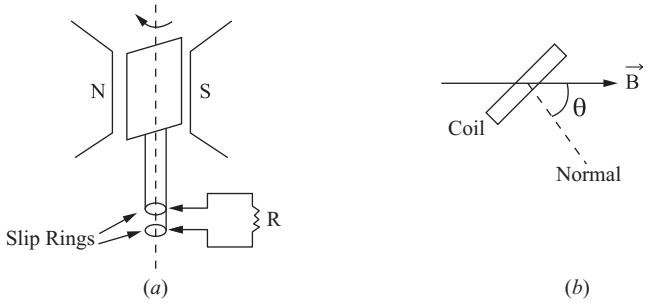


FIG. 13.1

The axis of rotation is at right angles to the field. As the coil rotates, the magnetic flux passing through it changes. Hence an emf is induced in the coil.

Suppose we start timing from the instant when the plane of the coil is at right angles to the field **B**, *i.e.*, when the angle between the normal to the plane of the coil and the direction of the field is zero. Then, at an instant t, the normal to the plane of the coil will make angle θ (= ωt) with the direction of **B**.

The magnetic flux linked with N turns of the coil is

$$\phi = NBA \cos\theta = NBA \cos\omega t.$$

where A (= ab) is the area of the coil.

The instantaneous induced emf,

$$E = -\frac{d\phi}{dt} = -\frac{d}{dt}(NBA \cos\omega t) = NBA\,\omega \sin\omega t = E_0 \sin\omega t$$

Here, $E_0 = NBA\omega$, called the *peak value* of the e.m.f.

Now, $\omega = 2\pi v$ where v = frequency of alternating voltage.

Thus, when $\omega t = 0$, $\sin\omega t = 0$ and $E = 0$

$\omega t = \pi/2$, $\sin\omega t = 1$ and $E = E_0$

$\omega t = \pi$, $\sin\omega t = 0$ and $E = 0$

$\omega t = 3\pi/2$, $\sin\omega t = -1$, and $E = -E_0$

and $\omega t = 2\pi$, $\sin\omega t = 0$ and $E = 0$ again,

A graph of E against ωt is a sine curve (Fig. 13.2). Such an e.m.f. is called an 'alternating e.m.f.' The resulting current in the coil, if the coil is part of a *closed circuit*, is the 'alternating current.'

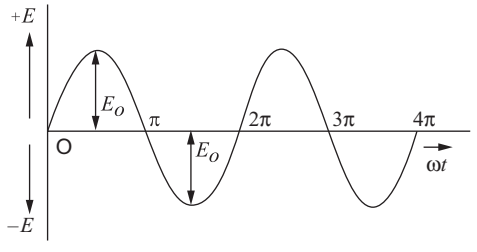


FIG. 13.2

The corresponding current I through the circuit is given by

$$I = I_0 \sin \omega t.$$

The time of one cycle is known as *time period T*, the number of cycles per second the frequency ν (= $1/T$), the peak value of current or voltage the *amplitude*.

Fig. 13.3 shows one cycle of ac voltage of current.

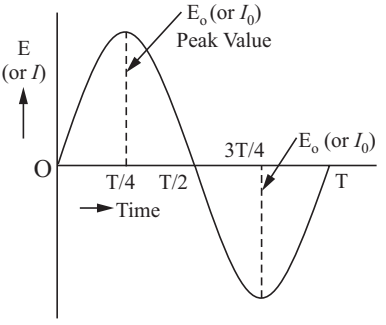


FIG. 13.3

Peak Value of Alternating Current or emf. *The maximum value of alternating current or emf in the positive or negative direction is called peak value of alternating current or emf. It is denoted by* I_0 or E_0.

Mean Value of Alternating Current. *Mean value of alternating current is defined as its average over half a cycle.*

$$I_{mean} = \frac{\int_0^{T/2} 1\, dt}{T/2} = \frac{\int_0^{\pi/\omega} I_0 \sin \omega t\, dt}{\pi/\omega}$$

$$= \frac{I_0 \omega}{\pi}\left[\frac{-\cos \omega t}{\omega}\right]_0^{\pi/\omega}$$

$$= -\frac{I_0}{\pi}[\cos \pi - \cos 0]$$

$$= \frac{2I_0}{\pi} = 0.637 I_0$$

Similarly, $E_{mean} = 0.637\, E_0$

Root mean square value of an alternating current. *It is defined as the square root of the average of* I^2 *during a complete cycle.*

$$\bar{I^2} = \frac{\int_0^{2\pi/\omega} I^2\, dt}{2\pi/\omega} = \frac{\int_0^{2\pi/\omega} I_0^2 \sin^2 \omega t\, dt}{2\pi/\omega}$$

$$= \frac{I_0^2\, \omega}{2\pi}\int_0^{2\pi/\omega} \frac{1}{2}(1 - \cos 2\omega t)\, dt$$

$$= \frac{I_0^2 \omega}{4\pi}\left[t - \frac{\sin 2\omega t}{2\omega}\right]_0^{2\pi/\omega}$$

$$= \frac{I_0^2 \omega}{4\pi}\left[\frac{2\pi}{\omega}\right] = \frac{I_0^2}{2}$$

$$I_{rms} = \sqrt{\bar{I^2}} = \frac{I_0}{\sqrt{2}} = 0.707\, I_0$$

Similarly, $E_{rms} = \frac{E_0}{\sqrt{2}} = 0.707\, E_0$

Form factor. The form factor gives an indication of the wave shape of the alternating voltage or current. *It is defined as the ratio of the virtual or r.m.s. value to the average value of alternating current or voltage.* Thus in the case of a sinusoidal current (or voltage), form factor is,

$$\text{Form factor} = \frac{I_{r.m.s}}{I_{mean}} = \frac{E_{r.m.s}}{E_{mean}} = \frac{0.707\, E_0}{0.637\, E_0} = 1.11$$

Effective value or virtual value of an A.C. *The rms value of an alternating current can also be defined as that direct current which produces the same rate of heating in a given resistance. Therefore, the r.m.s. value of alternating current is also called as the 'effective' or the 'virtual' value of the current.*

$$I_{virtual} = \frac{I_0}{\sqrt{2}} = I_{rms}$$

Suppose an alternating current of instantaneous value $I = I_0 \sin \omega t$ is flowing through a circuit of resistance R.

$$\left.\begin{matrix}\text{Total quantity of heat produced} \\ \text{over the complete cycle}\end{matrix}\right\} = H = \int_0^T I^2 \, R \, dt \qquad \ldots(1)$$

Let I_v stand for the root mean square or virtual value of the current. Then the heat produced in time T is given by

$$H = I_v^2 RT \qquad \ldots(2)$$

Comparing Eqs. (1) and (2),

$$I_v^2 \, RT = \int_0^T I^2 R \, dt \quad \text{or} \quad I_v^2 T = \int_0^T I^2 dt$$

or
$$I_v^2 T = \int_0^T I_0^2 \sin^2 \omega t \, dt = \frac{I_0^2}{2} \int_0^T 2 \sin^2 \omega t \, dt$$

$$= \frac{I_0^2}{2} \int_0^T (1 - \cos 2\omega t) \, dt = \frac{I_0^2}{2} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

$$= \frac{I_0^2 \cdot T}{2}$$

or
$$I_v^2 = \frac{I_0^2}{2}$$

or
$$I_v = \frac{I_0}{\sqrt{2}} = 0.707 \, I_0$$

Similarly, the r.m.s. value of an alternating voltage can be defined as that direct voltage which produces the same rate of heating, in a given resistance. The r.m.s. value of alternating voltage is also called as the 'effective' or the 'virtual' value of the voltage.

$$E_{virtual} = \frac{E_0}{\sqrt{2}} = E_{rms}$$

1. Mean or average value of A.C voltage for half cycle.

The alternating e.m.f is represented as,

$$E = E_0 \sin \omega t.$$

Mean value of alternating e.m.f over one half cycle is,

$$E_{av} = \frac{\int_0^{T/2} E_0 \sin \omega t \, dt}{\int_0^{T/2} dt} = \frac{E_0}{T/2} \left[\frac{-\cos \omega t}{\omega} \right]_0^{T/2}$$

$$= \frac{2E_0}{T} \times \frac{T}{2\pi} \left[-\cos \left(\frac{2\pi}{T} \right) t \right]_0^{T/2} \qquad \left(\because \quad \omega = \frac{2\pi}{T} \right)$$

$$= \frac{E_0}{\pi}[-\cos\pi + \cos 0]$$

$$\therefore \quad E_{av} = \frac{2E_0}{\pi}$$

2. R.M.S. value of Alternating Voltage

The R.M.S value of an alternating e.m.f is defined as that constant direct e.m.f which when applied to a resistor will develop the same heat energy in it as the given alternating e.m.f during the same interval of time.

The root mean square (r.m.s.) value of an alternating e.m.f is the square root of the mean of the squares of the emf taken during one cycle.

Instantaneous value of the e.m.f. = $E = E_0 \sin \omega t$

$$\text{Mean square value} = \frac{\int_0^T E_0^2 \sin^2 \omega t \, dt}{\int_0^T dt} = \frac{1}{T}\int_0^T E_0{}^2 \sin^2 \omega t \, dt$$

$$\int_0^T \sin^2 \omega t \, dt = \int_0^T \frac{1}{2}(1 - \cos 2\omega t)\, dt = \int_0^T \frac{1}{2}\, dt - \frac{1}{2}\int_0^T \cos 2\omega t \, dt$$

$$= \frac{1}{2}[t]_0^T - \frac{1}{2}\left[\frac{\sin 2\omega t}{2\omega}\right]_0^T = \frac{T}{2}$$

$$\text{Mean square value} = \frac{1}{T} \times E_0^2 \times \frac{T}{2} = \frac{E_0^2}{2}$$

$$\therefore \quad E_{rms} = E_0/\sqrt{2}$$

Impedance, Reactance and Admittance.

- Alternating current in a circuit is controlled by resistance, inductance and capacitance.

Impedance (Z). In alternating current circuit, the ratio of e.m.f. applied and consequent current produced is called the *impedance* and is denoted by Z

i.e., $$Z = \frac{E}{I}.$$

In any circuit, the ratio of the effective voltage to the effective current is defined as the impedance Z of the circuit.

Physically impedance of ac circuit is the hindrance offered by the circuit to the flow of ac through it.

Reactance (X): The hindrance offered by inductance and capacitance to the flow of ac in an ac circuit is called *reactance* and is denoted by X. Thus when there is no ohmic resistance in the circuit, the reactance is equal to impedance. The reactance due to inductance alone is called inductive reactance and is denoted by X_L. The reactance due to capacitance alone is called the capacitive reactance and is denoted by X_C.

Admittance. The inverse of impedance is called the admittance and is denoted by Y *i.e.*,

$$Y = \frac{1}{Z}$$